CatchFire Matching Engine -- Status Report

Date: February 26, 2026

Project: cf-influencer-matching-engine

Version: v0.8.0

Node.js: v24.11.1 | **Vulnerabilities:** 1 high (minimatch ReDoS in md-to-pdf, non-exploitable) | **Tests:** 119 standalone passing + 13 integration (require server)

System Health

SERVICE	STATUS	DETAILS
Cloud Run	Online	Revision cf-matching-engine-00018-hkm, us-central1
Firestore	Online	creators collection, 37 documents + scraper_runs collection
Gemini AI	Online	gemini-2.5-flash
Embeddings	Online	gemini-embedding-001, 768 dimensions
Golden Records	Active	11 benchmark creators across 8 crafts
GitHub CI	Passing	Unit tests + type check + build + smoke tests
Authentication	Active	Bearer token auth on all mutating endpoints
Python Scraper	Integrated	14 scrapers bundled in combined Docker container

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Executive Summary

The CatchFire Influencer Matching Engine is an AI-powered system for discovering and matching creators based on craft storytelling skills. The project has progressed through 8 development phases since January 28, 2026, and is now in a production-deployed state with a working scraper pipeline.

Current state:

- 24 API endpoints live in production on Cloud Run
- 10 web UI pages (React 19 + Vite 7 SPA)
- 37 creators in Firestore with 768-dimension embeddings
- 14 Python scrapers integrated into combined container (Feb 25)
- 119 standalone tests passing, CI/CD pipeline active
- 12 global Cursor rules governing AI workflow discipline
- 16 documented lessons learned in tasks/lessons.md

Accomplishments (Cumulative)

Phase 0-3: Foundation through Search (Jan 28)

- Express 5 API server with Zod validation
- Firestore creator database with full CRUD
- Gemini AI integration (categorize, style signatures, embeddings)
- Semantic search with 768-dimension embeddings

- Golden Record lookalike model (11 benchmark creators)
- Deployed to Cloud Run

Phase 4: Scraper Integration (Feb 12-25)

- Python scraper pipeline with 14 source scrapers (Ciclope, Camerimage, Motionographer, Annecy, UKMVA, The Rookies, ShotDeck, Directors Notes, Stash Media, Ars Electronica, Promax, Sitges, SXSW Title, Fantastic Fest)
- Schema transform module (3-tier Python output to BatchCreatorSchema)
- Combined container architecture (Node.js + Python in single Docker image)
- Scraper trigger endpoint with real orchestration via child_process
- Firestore scraper_runs collection for run tracking
- Batch import with auto-deduplication (name-normalized matching)
- Batch embedding generation (768d vectors)

Phase 5: Production Hardening (Feb 17-25)

- Rate limiting (express-rate-limit) on all public endpoints
- Helmet security headers
- Cloud Monitoring alert scripts (deploy-monitoring.sh)
- Cloud Scheduler job configs (daily Vimeo/Behance, weekly all-platform)
- Version bumped to v0.8.0 in health endpoint

Phase 6: Testing and QA (Feb 17-23)

- Vitest unit test suite: 45 schema + 41 scoring tests
- E2E semantic search tests: 33 tests across 7 suites
- Integration tests: 13 (require running server, see L016)
- Smoke test suite: 29/29 passing in production
- GitHub Actions CI/CD: unit tests + type check + build + smoke
- Test data fixtures: 12 mock creators, 768-dim embeddings

Phase 7: Web Application (Feb 18-25)

- React 19 + Vite 7 SPA with dark editorial luxury aesthetic
- 10 pages: Control Center, How It Works, Browse Creators, Creator Profile, Brief Match, Semantic Search, Admin, Status, Login, Scraper Dashboard
- ErrorBoundary + Toast notifications + ApiError class
- Brief Templates (10 pre-built across 5 categories)
- Bearer token authentication (Login page + middleware)
- /app root redirects to /creators

Infrastructure and DevOps (Feb 23)

- 2 new global Cursor rules (ai-workflow-discipline, task-management)
- Migrated task tracking to tasks/ convention (todo.md + lessons.md)
- 16 lessons documented (L001-L016)
- Root cause analysis of sprint loop, prevention rules deployed

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Database Statistics

METRIC	VALUE
Total Creators	37
Golden Records	11
Crafts Covered	8 (cinematographer, director, motion_designer, vfx_artist, colorist, animator, sound_designer, 3d_artist)
Platforms	5 (vimeo: 13, other: 18, behance: 3, artstation: 2, youtube: 1)
Embedding Dimensions	768
Scraper Sources	14

API Endpoints (24 Total)

Search and Discovery

- `POST /api/v1/search/semantic` -- AI-powered semantic search
- `GET /api/v1/lookalikes` -- Golden Record lookalike recommendations
- `GET /api/v1/similar/:id` -- Find creators similar to a given creator

Creator CRUD

- `GET /api/v1/creators` -- List/search/filter creators (optional auth)
- `GET /api/v1/creators/:id` -- Get creator by ID (optional auth)
- `POST /api/v1/creators` -- Add single creator (Zod validated, requires auth)
- `POST /api/v1/creators/batch` -- Bulk import (requires auth)
- `POST /api/v1/import/apify` -- Apify scraper import with LLM categorization (requires auth)

AI / LLM

- `POST /api/v1/match` -- Match creators to a brief
- `POST /api/v1/categorize` -- LLM auto-categorize from bio (requires auth)
- `POST /api/v1/style-signature` -- Generate style signature (requires auth)
- `GET /api/v1/llm/test` -- Test LLM connection (optional auth)
- `POST /api/v1/embed` -- Generate embeddings for a creator (requires auth)
- `POST /api/v1/embed/batch` -- Batch embedding generation (requires auth)

Scraper Management

- `POST /api/v1/scraper/trigger` -- Trigger scraper run with real orchestration (requires auth)
- `GET /api/v1/scraper/status` -- Current scraper status (requires auth)
- `GET /api/v1/scraper/reports` -- Recent scrape reports from Firestore (requires auth)

System

- `GET /health` -- Health check (public)
- `GET /api/v1/stats` -- Database statistics (optional auth)
- `POST /api/v1/feedback` -- Thumbs up/down feedback (requires auth)

Web UI

- `GET /` -- Beta Control Center (primary dashboard)
- `GET /dashboard` -- Analytics dashboard
- `GET /testing` -- Match testing UI

- `GET /app/*` -- React SPA (10 pages)

Test Coverage

SUITE	TESTS	STATUS
Unit -- schemas (vitest)	45	All passing
Unit -- scoring (vitest)	41	All passing
E2E -- semantic search (vitest)	33	All passing
Integration (vitest)	13	Require running local server (L016)
Smoke -- production	29	All passing (verified Feb 25)
Standalone Total	**119**	**All passing**

Security Posture

CONTROL	STATUS
Helmet security headers	Active
Rate limiting (express-rate-limit)	Active
CORS configuration	Active
Zod input validation	All POST endpoints
Bearer token authentication	All mutating endpoints
Scraper endpoint auth	Cloud Scheduler header or API key
npm audit	1 high (minimatch in md-to-pdf, non-exploitable)
No hardcoded secrets	All via .env

Deployment Info

FIELD	VALUE
Service	cf-matching-engine
Project	catchfire-app-2026
Region	us-central1
Revision	cf-matching-engine-00018-hkm
URL	https://cf-matching-engine-34240596768.us-central1.run.app
Last Deployed	2026-02-25
Architecture	Combined container (Node.js + Python 3)
Deploy Command	`gcloud run deploy cf-matching-engine --source . --region us-central1 --project catchfire-app-2026`

FIELD	VALUE
Env Vars	NODE_ENV, GCP_PROJECT, GCP_REGION, API_AUTH_TOKENS, SCRAPER_API_KEY

Critical deploy note: Always use `--update-env-vars` (additive), never `--set-env-vars` (replaces all, wipes auth tokens).

Known Issues

ISSUE	SEVERITY	NOTES
Integration tests need running server	Medium	L016 -- 13 tests require localhost:8090
Admin "Refresh Model" button fails without login	Low	Needs auth token in localStorage; visit /app/login first
No React SPA Home page	Low	/app redirects to /creators; proper landing page is future work
Cloud Scheduler not yet deployed	Medium	Job configs exist but not submitted to GCP
Staging env not yet deployed	Medium	Scripts + CI/CD ready, needs staging branch push
Scraper dashboard shows some mock data	Low	Reports endpoint uses Firestore scraper_runs but dashboard UI partially mocked
Login is token-only	Low	No OAuth/password flow yet
npm audit: minimatch ReDoS	Low	In md-to-pdf dep, not exploitable server-side

Next Steps

PRIORITY	TASK	NOTES
High	8.1 Image Analysis (Gemini Vision)	Auto-tag visual style from portfolios
High	8.3 Brief Templates	Pre-built search queries for common use cases
Medium	4.5 Scraping cadence scheduler	Festivals=annual, portfolios=quarterly, deploy Cloud Scheduler jobs
Medium	5.4 Staging environment	Deploy scripts ready, needs staging branch push
Medium	Admin login UX improvement	Auto-prompt or persist auth token
Medium	Wire scraper dashboard to real Firestore data	Replace remaining mock elements
Low	React SPA Home/landing page	Currently redirects to /creators
Low	Add loading skeletons to React pages	Replace "Loading..." text
Low	Fix 13 integration tests for CI	Need mock server or auto-start in test setup

Future Enhancements

PHASE	ENHANCEMENT	DESCRIPTION
8.1	Image Analysis (Gemini Vision)	Auto-tag visual style from creator portfolios using multimodal AI

PHASE	ENHANCEMENT	DESCRIPTION
8.2	Contact Enrichment (Clay.com/Hunter.io)	Enrich creator profiles with verified contact data ($1,000 budget allocated)
8.3	Brief Templates	Pre-built search queries across categories (Film, Brand, Motion, Music, Experiential)
8.4	Slack Integration	`/catchfire find` slash command for inline creator discovery
8.5	Auto-Categorize Improvements	Fine-tune LLM prompts based on match feedback data
8.6	Multi-Model Support	Gemini Pro for complex briefs, Flash for fast categorization
8.7	Multi-Chip Search (AND logic)	Click multiple hint chips to narrow search scope
8.8	Interactive Kanban Board	Real-time task tracking at /app/kanban with drag-and-drop
9.1	Automated Scraper Pipeline	End-to-end: scheduler fires, scraper runs, data imported, embeddings generated
9.2	Multi-User Auth	Role-based access (admin, viewer), Google OAuth SSO
9.3	Creator Analytics	Trend tracking, engagement metrics, portfolio freshness scoring
9.4	Brief History and Saved Searches	Save searches, view past briefs, share results
9.5	Webhook Notifications	Slack/email alerts for imports, scraper completions, match quality changes

Blocked / Waiting on Decision

ITEM	BLOCKED BY	RESOLUTION NEEDED
Golden Records expansion	Creative team input	Task 4.4 one-pager sent to Dan
Feedback sheet location	PM / Dan decision	Need FEEDBACK_SHEET_ID and tab name
External/cultural live data scope	Dan confirmation	Trend API needed or scraper+LLM sufficient?
Apify actor selection	Budget and ToS review	Which platforms to scrape first

File Structure (Key Paths)

PATH	PURPOSE
`src/index.js`	Main Express server with API endpoints
`src/schemas.ts`	Zod schemas and TypeScript types
`src/scoring.ts`	Scoring algorithm with typed creator/brief matching
`src/llm.ts`	Gemini AI integration
`src/routes/scraperTrigger.js`	Real scraper orchestration via child_process
`src/services/scraper-transform.js`	Schema transform (3-tier to BatchCreatorSchema)
`scraper/`	Python scraper pipeline (14 scrapers, pipeline, models)

PATH	PURPOSE
`tasks/todo.md`	Canonical task tracker
`tasks/lessons.md`	Known fixes and lessons learned (L001-L016)
`tests/*.test.ts`	Vitest unit, E2E, and integration tests
`web/`	React 19 + Vite 7 SPA source
`Dockerfile`	Combined container (Node.js + Python 3)

Global Cursor Rules (12 Total)

RULE	PURPOSE
ai-workflow-discipline.mdc	Plan mode, verification, elegance, autonomous bug fixing
task-management.mdc	tasks/ convention, self-improvement loop
session-wrap-up.mdc	End-of-session checklist and handoff
session-start-repo-sync.mdc	Repo sync check at session start
git-safety-archive.mdc	Archive remote before any git push
git-sync-workflow.mdc	Conflict resolution, latest-version preference
mandatory-plugins.mdc	claude-mem, code-simplifier, superpowers
global-learn.mdc	Cross-project LEARN-GLOBAL.md consultation
global-tools.mdc	Shared tools (session-pdf, etc.)
authorship-standards.mdc	AI co-authorship attribution on all files
auth-permissions.mdc	Pre-authorized credential usage
github-account-management.mdc	Account switching (cmscholz222 vs AbsoluteELEV8)

Recent Commits

DATE	COMMIT	CHANGE
Feb 25	`457fc7b`	Session handoff for scraper integration
Feb 25	`dd3a432`	Fix /app root redirect to /creators
Feb 25	`c0acf4e`	Integrate Python scraper pipeline into combined container
Feb 25	`9027d97`	Bump version to v0.8.0, deploy Cloud Monitoring alerts
Feb 23	`ccb5c60`	Workflow rules, tasks/ migration, stale file cleanup

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